

Reverse-check review package — NJLComposite_gen

An independent, **blank-slate** agent instance reconstructed the physics below from the sanitized `.fr` alone (no paper, no metadata, no conversation history). A second fresh instance then compared the reconstruction against the source paper. The final verdict belongs to the human reviewer — sign off at the bottom.

item	value
model file	NJLComposite/model/NJLComposite_gen.fr
original model name	NJLComposite_gen (hidden from the agent)
paper	NJLComposite/text/2311.18472.txt
blindness scope	blank slate w.r.t. paper, authors, model name, comments and prose labels — NOT w.r.t. field/parameter symbol names, which are kept so the reviewer can map the reconstruction back

Verbatim Lagrangian terms (from the `.fr`)

These are the terms the reconstruction must account for, quoted unmodified. Check each against its reconstructed LaTeX form below.

LkinNJL (=)

```
Block[{mu, aa}, ExpandIndices[DC[HC[Pi5e[aa]], mu] DC[Pi5e[aa], mu] - MPi5e^2 HC[Pi5e[aa]]
  ↳ Pi5e[aa] + DC[HC[Pi5mu[aa]], mu] DC[Pi5mu[aa], mu] - MPi5mu^2 HC[Pi5mu[aa]] Pi5mu[aa] +
  ↳ DC[HC[Pi5tau[aa]], mu] DC[Pi5tau[aa], mu] - MPi5tau^2 HC[Pi5tau[aa]] Pi5tau[aa] +
  ↳ DC[HC[Pi2de[aa]], mu] DC[Pi2de[aa], mu] - MPi2de^2 HC[Pi2de[aa]] Pi2de[aa] +
  ↳ DC[HC[Pi2dmu[aa]], mu] DC[Pi2dmu[aa], mu] - MPi2dmu^2 HC[Pi2dmu[aa]] Pi2dmu[aa] +
  ↳ DC[HC[Pi2dtau[aa]], mu] DC[Pi2dtau[aa], mu] - MPi2dtau^2 HC[Pi2dtau[aa]] Pi2dtau[aa] +
  ↳ DC[HC[Pi2ue[aa]], mu] DC[Pi2ue[aa], mu] - MPi2ue^2 HC[Pi2ue[aa]] Pi2ue[aa] +
  ↳ DC[HC[Pi2umu[aa]], mu] DC[Pi2umu[aa], mu] - MPi2umu^2 HC[Pi2umu[aa]] Pi2umu[aa] +
  ↳ DC[HC[Pi2utau[aa]], mu] DC[Pi2utau[aa], mu] - MPi2utau^2 HC[Pi2utau[aa]] Pi2utau[aa] +
  ↳ DC[HC[Pim1e[aa]], mu] DC[Pim1e[aa], mu] - MPim1e^2 HC[Pim1e[aa]] Pim1e[aa] +
  ↳ DC[HC[Pim1mu[aa]], mu] DC[Pim1mu[aa], mu] - MPim1mu^2 HC[Pim1mu[aa]] Pim1mu[aa] +
  ↳ DC[HC[Pim1tau[aa]], mu] DC[Pim1tau[aa], mu] - MPim1tau^2 HC[Pim1tau[aa]] Pim1tau[aa]]]
```

LNJLYukawaNonHC (:=)

```
Block[{sp, aa}, ExpandIndices[lamEU anti[lR][sp, 1].QL[sp, 1, 1, aa] HC[Pi5e][aa] + lamMuC
  ↳ anti[lR][sp, 2].QL[sp, 1, 2, aa] HC[Pi5mu][aa] + lamTauT anti[lR][sp, 3].QL[sp, 1, 3,
  ↳ aa] HC[Pi5tau][aa] + lamED anti[lR][sp, 1].QL[sp, 2, 1, aa] HC[Pi2de][aa] + lamMuS
  ↳ anti[lR][sp, 2].QL[sp, 2, 2, aa] HC[Pi2dmu][aa] + lamTauB anti[lR][sp, 3].QL[sp, 2, 3,
  ↳ aa] HC[Pi2dtau][aa] + lamNuEU anti[LL][sp, 1, 1].QL[sp, 1, 1, aa] HC[Pi2ue][aa] +
  ↳ lamNuMuC anti[LL][sp, 1, 2].QL[sp, 1, 2, aa] HC[Pi2umu][aa] + lamNuTauT anti[LL][sp, 1,
  ↳ 3].QL[sp, 1, 3, aa] HC[Pi2utau][aa] + lamNuED anti[LL][sp, 1, 1].QL[sp, 2, 1, aa]
  ↳ HC[Pim1e][aa] + lamNuMuS anti[LL][sp, 1, 2].QL[sp, 2, 2, aa] HC[Pim1mu][aa] + lamNuTauB
  ↳ anti[LL][sp, 1, 3].QL[sp, 2, 3, aa] HC[Pim1tau][aa], FlavorExpand -> {Generation}]]
```

LNJLYukawa (:=)

LNJLYukawaNonHC + HC[LNJLYukawaNonHC]

LBSM (=)

LkinNJL + LNJLYukawa

Blank-slate reconstruction

Reconstruction of `sanitized.fr`

Lagrangian

Notation: color indices are $a, b = 1, 2, 3$. The SM fermions are written as

$$Q_{Li}^a = (u_{Li}^a, d_{Li}^a)^T, \quad L_{Li} = (\nu_{Li}, \ell_{Li})^T,$$

with $i = 1, 2, 3$ for $e/\mu/\tau$ -aligned generations. Thus $u_L, c_L, t_L, d_L, s_L, b_L, e_R, \mu_R, \tau_R$, and $\nu_{eL}, \nu_{\mu L}, \nu_{\tau L}$ appear.

For each new scalar ϕ^a , the kinetic operator uses the FeynRules `DC` covariant derivative. From the declarations, all new scalars carry a color index and no explicit weak-isospin index. The gauge content encoded in the file is therefore

$$D_\mu \phi^a = \partial_\mu \phi^a - ig_s G_\mu^A (T^A)^a_b \phi^b + \text{neutral electroweak couplings fixed by the declared } Q \text{ and } Y.$$

Equivalently, in a broken-electroweak notation,

$$D_\mu \phi^a = \left[\partial_\mu - ig_s G_\mu^A T^A - ie Q_\phi A_\mu - i \frac{e}{s_W c_W} (T_{3,\phi} - Q_\phi s_W^2) Z_\mu \right]^a_b \phi^b, \quad T_{3,\phi} = Q_\phi - Y_\phi.$$

No charged $W^\pm$ covariant-derivative mixing between the component fields is explicitly declared because the fields have no `SU2W` index.

`LkinNJL`

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{5e}^a)^\dagger (D^\mu \Pi_{5e}^a) - M_{\Pi 5e}^2 \Pi_{5e}^{a\dagger} \Pi_{5e}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{5\mu}^a)^\dagger (D^\mu \Pi_{5\mu}^a) - M_{\Pi 5\mu}^2 \Pi_{5\mu}^{a\dagger} \Pi_{5\mu}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{5\tau}^a)^\dagger (D^\mu \Pi_{5\tau}^a) - M_{\Pi 5\tau}^2 \Pi_{5\tau}^{a\dagger} \Pi_{5\tau}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2de}^a)^\dagger (D^\mu \Pi_{2de}^a) - M_{\Pi 2de}^2 \Pi_{2de}^{a\dagger} \Pi_{2de}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2d\mu}^a)^\dagger (D^\mu \Pi_{2d\mu}^a) - M_{\Pi 2d\mu}^2 \Pi_{2d\mu}^{a\dagger} \Pi_{2d\mu}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2d\tau}^a)^\dagger (D^\mu \Pi_{2d\tau}^a) - M_{\Pi 2d\tau}^2 \Pi_{2d\tau}^{a\dagger} \Pi_{2d\tau}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2ue}^a)^\dagger (D^\mu \Pi_{2ue}^a) - M_{\Pi 2ue}^2 \Pi_{2ue}^{a\dagger} \Pi_{2ue}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2u\mu}^a)^\dagger (D^\mu \Pi_{2u\mu}^a) - M_{\Pi 2u\mu}^2 \Pi_{2u\mu}^{a\dagger} \Pi_{2u\mu}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{2u\tau}^a)^\dagger (D^\mu \Pi_{2u\tau}^a) - M_{\Pi 2u\tau}^2 \Pi_{2u\tau}^{a\dagger} \Pi_{2u\tau}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{-1e}^a)^\dagger (D^\mu \Pi_{-1e}^a) - M_{\Pi -1e}^2 \Pi_{-1e}^{a\dagger} \Pi_{-1e}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{-1\mu}^a)^\dagger (D^\mu \Pi_{-1\mu}^a) - M_{\Pi -1\mu}^2 \Pi_{-1\mu}^{a\dagger} \Pi_{-1\mu}^a$$

$$\mathcal{L}_{\text{LkinNJL}} \supset (D_\mu \Pi_{-1\tau}^a)^\dagger (D^\mu \Pi_{-1\tau}^a) - M_{\Pi -1\tau}^2 \Pi_{-1\tau}^{a\dagger} \Pi_{-1\tau}^a$$

LNJLYukawaNonHC

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{EU} \bar{e}_R u_L^a \Pi_{5e}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\mu C} \bar{\mu}_R c_L^a \Pi_{5\mu}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\tau T} \bar{\tau}_R t_L^a \Pi_{5\tau}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{ED} \bar{e}_R d_L^a \Pi_{2de}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\mu S} \bar{\mu}_R s_L^a \Pi_{2d\mu}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\tau B} \bar{\tau}_R b_L^a \Pi_{2d\tau}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu EU} \bar{\nu}_{eL} u_L^a \Pi_{2ue}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu\mu C} \bar{\nu}_{\mu L} c_L^a \Pi_{2u\mu}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu\tau T} \bar{\nu}_{\tau L} t_L^a \Pi_{2u\tau}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu ED} \bar{\nu}_{eL} d_L^a \Pi_{-1e}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu\mu S} \bar{\nu}_{\mu L} s_L^a \Pi_{-1\mu}^{a\dagger}$$

$$\mathcal{L}_{\text{LNJLYukawaNonHC}} \supset \lambda_{\nu\tau B} \bar{\nu}_{\tau L} b_L^a \Pi_{-1\tau}^{a\dagger}$$

LNJLYukawa

$$\mathcal{L}_{\text{LNJLYukawa}} = \mathcal{L}_{\text{LNJLYukawaNonHC}} + \mathcal{L}_{\text{LNJLYukawaNonHC}}^\dagger$$

LBSM

$$\mathcal{L}_{\text{LBSM}} = \mathcal{L}_{\text{LkinNJL}} + \mathcal{L}_{\text{LNJLYukawa}}$$

Field Table

.fr symbol	Spin	SU(3) rep	SU(2) rep declared	Q	Y	Self- conjugate	Mass
Pi5e	0	3	none / singlet as declared	5/3	7/6	no	MPi5e = 1000.
Pi5mu	0	3	none / singlet as declared	5/3	7/6	no	MPi5mu = 1000.

.fr symbol	Spin	SU(3) rep	SU(2) rep declared	Q	Y	Self- conjugate	Mass
Pi5tau	0	3	none / singlet as declared	5/3	7/6	no	MPi5tau = 1000.
Pi2de	0	3	none / singlet as declared	2/3	7/6	no	MPi2de = 1000.
Pi2dmu	0	3	none / singlet as declared	2/3	7/6	no	MPi2dmu = 1000.
Pi2dtau	0	3	none / singlet as declared	2/3	7/6	no	MPi2dtau = 1000.
Pi2ue	0	3	none / singlet as declared	2/3	1/6	no	MPi2ue = 1000.
Pi2umu	0	3	none / singlet as declared	2/3	1/6	no	MPi2umu = 1000.
Pi2utau	0	3	none / singlet as declared	2/3	1/6	no	MPi2utau = 1000.
Pim1e	0	3	none / singlet as declared	-1/3	1/6	no	MPim1e = 1000.
Pim1mu	0	3	none / singlet as declared	-1/3	1/6	no	MPim1mu = 1000.
Pim1tau	0	3	none / singlet as declared	-1/3	1/6	no	MPim1tau = 1000.

All new scalar fields also declare

$$B = 1/3, \quad L = -1,$$

and widths $W \dots = 1..$

Parameters

Parameter	Value	Multiplies	Physical meaning
1amEU	1.0	$\bar{e}_R u_L^a \Pi_{5e}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amMuC	1.0	$\bar{\mu}_R c_L^a \Pi_{5\mu}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amTauT	1.0	$\bar{\tau}_R t_L^a \Pi_{5\tau}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amED	1.0	$\bar{e}_R d_L^a \Pi_{2de}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amMuS	1.0	$\bar{\mu}_R s_L^a \Pi_{2d\mu}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amTauB	1.0	$\bar{\tau}_R b_L^a \Pi_{2d\tau}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling

Parameter	Value	Multiplies	Physical meaning
1amNuEU	1.0	$\bar{\nu}_{eL} u_L^a \Pi_{2ue}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amNuMuC	1.0	$\bar{\nu}_{\mu L} c_L^a \Pi_{2u\mu}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amNuTauT	1.0	$\bar{\nu}_{\tau L} t_L^a \Pi_{2u\tau}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amNuED	1.0	$\bar{\nu}_{eL} d_L^a \Pi_{-1e}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amNuMuS	1.0	$\bar{\nu}_{\mu L} s_L^a \Pi_{-1\mu}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling
1amNuTauB	1.0	$\bar{\nu}_{\tau L} b_L^a \Pi_{-1\tau}^{a\dagger}$	real Yukawa-like scalar leptoquark coupling

Physics Summary

The file encodes twelve complex color-triplet scalar states with baryon number 1/3 and lepton number -1 , each coupled diagonally to one lepton generation and one quark generation. The interactions are scalar leptoquark-like Yukawa couplings involving $\bar{\ell}_R q_L$ or $\bar{\nu}_L q_L$, plus Hermitian conjugates.

These states mediate quark-lepton transitions such as $q\ell \leftrightarrow \Pi$, scalar pair production through QCD gauge interactions, and t - or s -channel quark-lepton scattering with generation-aligned final states.

Paper cross-check

Comparison of `reconstruction.md` Against the Paper Model

Paper Model Definition Located

The paper defines the relevant composite leptoquark model in **Section II**, “Nambu-Jona-Lasinio Composite Leptoquark Model.”

Key references:

- **Table I**: composite boson field content, constituents, electric charge $Q_i = Y + t_{3L}^i$, weak isospin t_{3L}^i , hypercharge Y , color representation, and LQ nomenclature.
- **Eq. (1)**: scalar leptoquark kinetic and mass term,

$$(D_\mu \Phi)^\dagger (D^\mu \Phi) + M_\Pi^2 \Phi^\dagger \Phi.$$

- **Eq. (2)**: covariant derivative,

$$D_\mu = \partial_\mu + ig_1 Y B_\mu + \frac{1}{2} ig_2 \sigma^i W_\mu^i + ig_3 T^a G_\mu^a.$$

- Text immediately after **Eq. (2)**: the color-triplet LQ bosons form $SU_L(2)$ doublets,

$$\Pi_{1/6}^a = \begin{pmatrix} \Pi_{ua}^{2/3} \\ \Pi_a^{-1/3} \end{pmatrix}, \quad \Pi_{7/6}^a = \begin{pmatrix} \Pi_a^{5/3} \\ \Pi_{da}^{2/3} \end{pmatrix}.$$

- **Eq. (3)**: effective contact/Yukawa interactions for the first generation, plus h.c.
- Text after **Eq. (3)**: generalization to second and third generations, baryon/lepton numbers $B = 1/3$, $L = -1$, and conjugation relations.
- **Eq. (4)**: example flavor-mixed Yukawa couplings, with CKM-like factors.
- **Section III**: states that the MadGraph/FeynRules implementation includes the gauge terms of Eq. (1) and contact terms of Eq. (3), implemented up to the third fermion generation.

Term-by-Term Comparison

paper term (eq. ref)	reconstruction term	verdict	notes
Table I and text after Eq. (2): $SU_c(3)$ triplet scalar LQs arranged as $SU_L(2)$ doublets $\Pi_{7/6}^a = (\Pi_a^{5/3}, \Pi_{da}^{2/3})^T$, $\Pi_{1/6}^a = (\Pi_{ua}^{2/3}, \Pi_a^{-1/3})^T$.	Twelve complex color-triplet scalars, but each has “SU(2) rep declared: none / singlet as declared”; no explicit weak-isospin index.	disagree	Component electric charges and hypercharges match the two paper doublets, but the reconstruction says the implementation has no $SU(2)$ doublet structure. This removes explicit charged- $W^\pm$ transitions between doublet components required by Eq. (2).
Eq. (1): $(D_\mu \Phi)^\dagger (D^\mu \Phi) + M_\Pi^2 \Phi^\dagger \Phi$.	For each scalar, $(D_\mu \Pi)^\dagger (D^\mu \Pi) - M_\Pi^2 \Pi^\dagger \Pi$.	disagree	The kinetic form agrees, but the mass term sign differs from the paper as written. This may be a metric/sign convention or a paper typo, but as text-to-text comparison it is different.
Eq. (2): covariant derivative includes $+\frac{1}{2}ig_2\sigma^i W_\mu^i$, acting on $SU(2)$ doublets.	Covariant derivative includes QCD and neutral electroweak couplings fixed by Q, Y , with no charged $W^\pm$ covariant-derivative mixing.	disagree	This is the same core issue as the field representation disagreement: the paper has Pauli-matrix $SU(2)$ action on doublets; the reconstruction describes component singlets with no $W^\pm$ doublet mixing.
Table I: $\Pi_a^{5/3} \propto \bar{e}_R u_{La}$, $Q = 5/3$, $t_3 = +1/2$, $Y = 7/6$, color triplet; Eq. (3): $g_{\Pi 5/3}(\bar{e}_R u_{La})\Pi_a^{-5/3} + \text{h.c.}$.	$\lambda_{EU}\bar{e}_R u_L^a \Pi_{5e}^{a\dagger}$, plus μc and τt analogues.	agree	Chirality, electric charge, hypercharge, color, conjugation in the interaction, and generation generalization agree. Reconstruction uses separate generation labels and default numerical couplings.
Table I: $\Pi_{da}^{2/3} \propto \bar{e}_R d_{La}$, $Q = 2/3$, $t_3 = -1/2$, $Y = 7/6$; Eq. (3): $g_{\Pi -2/3}(\bar{e}_R d_{La})\Pi_{da}^{-2/3} + \text{h.c.}$.	$\lambda_{ED}\bar{e}_R d_L^a \Pi_{2de}^{a\dagger}$, plus μs and τb analogues.	agree	The component assignment, chirality, color, charge, hypercharge, and use of the conjugate scalar in the interaction agree.

paper term (eq. ref)	reconstruction term	verdict	notes
Table I: $\Pi_{ua}^{2/3} \propto \bar{\nu}_e^R u_{La}$, $Q = 2/3$, $t_3 = +1/2$, $Y = 1/6$; Eq. (3): $g_{\Pi-2/3}(\bar{\nu}_e^R u_{La})\Pi_{ua}^{-2/3} +$ h.c..	$\lambda_{\nu EU}\bar{\nu}_e u_L^a \Pi_{2ue}^{a\dagger}$, plus $\nu_\mu c$ and $\nu_\tau t$ analogues.	disagree	The scalar charge and hypercharge match, and conjugation is analogous, but the paper uses a barred right-handed neutrino constituent in Table I/Eq. (3), while the reconstruction uses left-handed neutrinos.
Table I: $\Pi_a^{-1/3} \propto \bar{\nu}_e^R d_{La}$, $Q = -1/3$, $t_3 = -1/2$, $Y = 1/6$; Eq. (3): $g_{\Pi 1/3}(\bar{\nu}_e^R d_{La})\Pi_a^{1/3} + \text{h.c.}$.	$\lambda_{\nu ED}\bar{\nu}_e d_L^a \Pi_{-1e}^{a\dagger}$, plus $\nu_\mu s$ and $\nu_\tau b$ analogues.	disagree	Scalar charge, hypercharge, color, and conjugation match, but neutrino chirality differs from the paper's Table I/Eq. (3).
Text after Eq. (3): conjugated fields $(\Pi^{5/3})^\dagger = \Pi^{-5/3}$, $(\Pi^{-1/3})^\dagger = \Pi^{1/3}$, and conjugates for the $Q = 2/3$ states.	All new scalars are complex and non-self-conjugate; Yukawa Lagrangian includes h.c.	agree	The reconstruction does not introduce separate independent antiparticle fields, but the complex scalar plus h.c. treatment captures the conjugate interactions.
Text after Eq. (3): $B = 1/3$, $L = -1$ for these LQ states.	All new scalar fields declare $B = 1/3$, $L = -1$.	agree	Matches the paper.
Text after Eq. (3): spectra and interactions generalized by $\nu_e, e, u, d \rightarrow \nu_\mu, \mu, c, s$ and ν_τ, τ, t, b .	Reconstruction includes diagonal first-, second-, and third-generation couplings.	agree	The diagonal generation generalization is represented.
Eq. (3): $g_{\Pi i} = (F_{\Pi i}/\Lambda)^2 \sim O(1)$; paper later uses λ for $g_{\Pi i}$.	Independent real parameters λ_{EU} , $\lambda_{\mu C}$, etc., defaulting to 1.0.	agree	The coefficient structure is compatible with the paper's $\lambda \sim O(1)$ convention. Equal default values are benchmark choices, not a structural disagreement.
Text after Eq. (3): masses $M_{\Pi i}$ and couplings $g_{\Pi i}$ can differ across generations.	Separate mass symbols for each scalar, but all default to 1000; separate coupling symbols, all default to 1.0.	agree	The reconstruction has independent parameters even though the displayed defaults are common benchmark values.
Eq. (4): example flavor-mixed couplings such as $g_{\Pi 5/3}(U_R^\dagger U_L)_{1,2}(\bar{\mu}_R c_L)\Pi_a^{-5/3}$ and $(U_R^\dagger U_L)_{1,3}(\bar{\tau}_R t_L)\Pi_a^{-5/3}$, with analogous terms possible for other LQs.	Only generation-aligned couplings are listed: eu , μc , τt , ed , μs , τb , and neutrino analogues.	missing-in-reconstruction	Eq. (4) describes flavor-mixed couplings of a given LQ state to other generations via mixing matrices. The reconstruction instead has separate generation-labeled LQ fields with diagonal couplings and no explicit CKM-like mixing factors.

paper term (eq. ref)	reconstruction term	verdict	notes
Section III: implementation includes gauge Eq. (1) and contact Eq. (3) terms up to third generation.	LBSM = LkinNJL + LNJLYukawa, with twelve scalar fields and Yukawa terms plus h.c.	agree	At the broad implementation-scope level, the reconstruction matches the stated UFO contents, except for the specific disagreements above.

Disagreements and Human Checks

1. **Missing $SU(2)_L$ doublet structure and charged- W interactions — substantive.**
A human should check whether the actual implementation intentionally broke the paper’s Eq. (2) doublet structure into electroweak-component singlets, because this changes gauge interactions involving $W^\pm$.
2. **Neutrino chirality in the $Y = 1/6$ leptoquark interactions — substantive.**
A human should check the implementation file’s chiral projectors for the neutrino vertices against Table I and Eq. (3), since the paper uses $\bar{\nu}_R q_L$ for the positive-charge $Y = 1/6$ constituents while the reconstruction reports $\bar{\nu}_L q_L$.
3. **Mass-term sign relative to Eq. (1) — convention.**
A human should check the paper’s metric/sign convention and the FeynRules scalar mass convention before treating this as a physical mismatch.
4. **Flavor-mixed Yukawa terms of Eq. (4) absent from the reconstruction — substantive.**
A human should check whether Eq. (4) was intended as part of the implemented model or only as a phenomenological possibility, because the reconstruction contains only generation-aligned couplings without explicit mixing-matrix factors.

Overall Assessment

The reconstruction captures the paper’s broad leptoquark content: complex color-triplet scalar states with charges $5/3$, $2/3$, and $-1/3$, baryon number $1/3$, lepton number -1 , generation-aligned Yukawa-like couplings, and hermitian conjugates. The main mismatches are notationally small but physically important: the paper defines the charged states as components of $SU(2)_L$ doublets with the full W_μ^i covariant derivative, while the reconstruction describes separate weak singlets with no charged- W mixing; and the neutrino-coupled $Y = 1/6$ terms use left-handed neutrinos in the reconstruction where the paper’s Table I and Eq. (3) use right-handed neutrino constituents. The absence of Eq. (4)-type flavor-mixed couplings is also important if the review target is the full paper model rather than only a simplified diagonal UFO benchmark.

Suggested checks for the reviewer

1. Every verbatim `.fr` term above has a reconstructed LaTeX counterpart with the same field content, chirality and conjugation.
2. Kinetic terms: covariant derivative gauge content matches the field’s representations; normalization is canonical.
3. Non-self-conjugate interaction terms appear together with their Hermitian conjugates.
4. Quantum numbers in the field table match the `.fr` declarations (and the paper, where the cross-check table flags disagreements).
5. Numeric masses/couplings are placeholders unless the paper pins them — treat values as demo inputs, not measurements.
6. Sanitizer scope: 24 prose labels were scrubbed; field/parameter symbol names were kept and may hint at the model’s identity.

Physicist sign-off

- Reviewed by: _____ Date: _____

- Verdict: ☐ approve ☐ approve with corrections ☐ reject
- Notes: